

Run R184 Metric Selection Note

Research reading artifact | fictional/internal | 2026-07-13

Analytical frame. For the fictional Helix Forge Lab snapshot dated 2026-07-13, run R184 compares M-14 at macro F1 0.873 with B-07 at 0.846. HF-Eval-S1/S2 v3.2 contains exactly 2,400 synthetic records: 1,600 S1 and 800 S2, divided among five labels.

Decision criteria

The lead metric must give all five synthetic labels a voice, remain understandable in review, and expose a regression that accuracy could conceal.

Metric failure cases

A high aggregate score can hide abstain failures; precision alone can reward withholding; recall alone can reward indiscriminate assignment.

Gate recommendation

Rank candidates by macro F1, then require no label-level collapse and inspect precision-recall movement before accepting a change.

Working matrix

Candidate	Useful when	Failure to watch
Macro F1	Labels are co-equal	Averages can hide spread.
Precision	False assignments matter	May reward abstention.
Recall	Misses matter	May reward over-routing.

Scored metric decision matrix

Metric	Label voice	Auditability	Misuse risk	Decision
Macro F1	High	High	Medium	Primary rank
Precision	Medium	High	High	Guardrail
Recall	Medium	High	High	Guardrail
Accuracy	Low	Medium	High	Context only

Gate checklist for the R184 reading

- Confirm each of the five synthetic labels has a visible F1.
- Reject a headline improvement that conceals a label-level collapse.
- Record whether precision and recall move in the expected direction.
- Attach the comparison to *Metric Gate Memorandum* and *R184 Comparator Worksheet* [synthetic/internal].

Recommended gate record

Decision: retain macro F1 as the rank metric, with precision and recall as mandatory counter-readings. A candidate passes only when its five-label scorecard is attached to the R184 review.